

The Treasury Lock

A comprehensive study

Part I – Practitioner Pricing Model — Part II – Hedging and Convexity Analysis (Pucci 2019)

Working notes

Source papers:

(1) Treasury Lock Model, *anonymous practitioner note (no author, no date)*.

(2) M. Pucci, Hedging the Treasury Lock, Banca IMI SpA, Milan, August 19, 2019, SSRN no. 3386521, <https://ssrn.com/abstract=3386521>.

Contents

1	Introduction and Trade Description	2
2	Unified Notation	3
3	The Bond Forward Pricing Building Block	4
4	The Three Lock Conventions	4
4.1	Lock the clean price ($P^{\text{T-Lock}}$ struck)	4
4.2	Lock the yield ($y^{\text{T-Lock}}$ struck)	5
4.3	Lock the dirty price	5
5	Risks and Sensitivities (Practitioner View)	5
6	The Arbitrage-Free Pricing Formula	6
7	The Forward-Contract Proxy and Overhedge	6
7.1	The proxy strike	6
7.2	The overhedge error is the Taylor remainder	7
8	Greeks of the T-Lock	7
8.1	Derivatives of the bond price	7
8.2	Payoff derivatives	7
8.3	Delta of the T-Lock	8
8.4	Gamma of the T-Lock	8
8.5	Sharp negativity bounds	8
9	Replication via Repo	9
10	The Then-Current T-Lock and Roll P&L	9
10.1	The roll P&L formula	9
10.2	Forming the new benchmark	11
10.3	Strike gap for the Then-Current proxy	11

11 Statistical Validation (Pucci 2019)	11
11.1 IRR descriptive statistics (since Jan-2008)	11
11.2 Overhedge error (6M T-Lock on 10Y)	11
11.3 Upper bound for negative gamma — eq. (27)	12
11.4 Roll P&L on the 10Y (trader’s view, 3% locked yield)	12
11.5 Roll P&L on the 5Y (trader’s view, 3%)	12
11.6 Roll P&L on the 2Y (trader’s view, 3%)	12
12 Valuation Adjustments	12
13 Conclusions (Pucci 2019)	13
14 Validation and Numerical Examples	13
14.1 Pucci’s worked example — 24 January 2019	13
14.2 Library-ready parameter block (pseudocode)	14
14.3 Cross-formula consistency	14
14.4 Risk: what to compute and how to interpret	14
14.5 Smoke tests	15
A Simple Math: Chain Rule and Inverse Function	15
B Key Derivatives of the Bond Price	15
C More Math: The Convexity-Inequality Proof	15
D The Forward Price and the Repo Strategy	16
D.1 The repo-cash strategy (zero haircut)	17
D.2 Unsecured-funding limit and the blend	17
E References	18

1 Introduction and Trade Description

A *Treasury Lock* (T-Lock) is a customised over-the-counter agreement that fixes either the *yield* or the *price* of a specified Treasury security for a specific period. The buyer is protected from a rise in the yield (equivalently a fall in price) of the underlying Treasury during the lock period. The lock period is normally one week to twelve months, most often ≤ 6 months.

T-Locks are most commonly used as a *pre-hedge* for an upcoming USD bond or loan issuance: the driver of the fixed rate paid by the borrower is the sum of a credit spread and the benchmark Treasury yield, so the client locks the Treasury component a few months ahead of issue.

Being **long** a T-Lock at strike L (the Locked Yield) means receiving at Expiry T_{del} an amount

$$g = a \cdot N \cdot \text{RF}_{T_{\text{del}}} \cdot (\text{IRR}_{T_{\text{del}}} - L), \quad a = +1 \text{ (long)}, -1 \text{ (short)}, \quad (1)$$

where

- N is the notional;
- $\text{IRR}_{T_{\text{del}}}$ is the bond’s internal rate of return at expiry;
- $\text{RF}_t = -\partial P_t(y)/\partial y|_{y=\text{IRR}_t}$ is minus the derivative of price w.r.t. yield (so $\text{RF} = -10,000 \cdot \text{DV01}$).
Two contract variants exist:
- **Standard T-Lock:** the underlying Treasury is specified at trade inception.
- **Then-Current T-Lock:** the underlying is the on-the-run n -Year benchmark prevailing at T_{del} ($n = 1, \dots, 30$; typically $n = 5, 7, 10$).

Hedge accounting. The T-Lock can be designated as the hedging instrument in a cash-flow hedge of an anticipated borrowing transaction. P&L is set aside in OCI until expiry; if the borrowing materialises the P&L is released amortised over the bond’s life, partially offsetting debt-service cash flows. If financial close fails, the full P&L recycles as cost/income at expiry (IFRS 9).

This document brings together two complementary perspectives:

Part I — Practitioner Pricing Model (anonymous note). Gives the operational pricing formulas a trading desk needs: the bond forward price with explicit coupon stripping (eq. (3)), and three NPV formulas for the three lock conventions (clean-price lock, yield lock, dirty-price lock).

Part II — Hedging and Convexity Analysis (Pucci 2019). Justifies the desk practice of *booking* a (short) T-Lock as a (long) forward contract on the same security. Shows the resulting overhedge is (i) bounded analytically by a Taylor remainder, (ii) generates non-negative gamma P&L (negative-convexity overhedge), (iii) statistically negligible at typical $\leq 6\text{M}$ tenors with at-the-money strikes.

2 Unified Notation

Symbol	Meaning
t, T_0	Valuation date / spot settlement date
T_{del}, t_e	Delivery date = T-Lock Expiry (interchangeable)
P_0	Spot clean price of the underlying bond
A_0, A_{del}	Accrued interest at T_0 and at T_{del}
C_i, T_i	i -th coupon and its payment date, $T_0 < T_i \leq T_{\text{del}}$
c	Coupon rate of the bond (continuous-compounding convention, Part II)
α_i	Accrual factor for the i -th coupon period
N	Number of coupons paid in $(T_0, T_{\text{del}}]$ (Part I) / bond notional (Part II)
M	Trade amount for a yield-locked T-Lock (Part I, eq. (6))
$P_t^{\text{mkt}}, P_t(y)$	Market dirty price / price as a function of yield
IRR_t	Internal rate of return at t : solution of $P_t^{\text{mkt}} = P_t(\text{IRR}_t)$
B_f	Bond forward (clean) price at delivery
y_f	Forward bond yield (= the IRR implied by B_f)
$L, P^{\text{T-Lock}}, y^{\text{T-Lock}}$	Locked yield / locked clean price / locked yield (strike of the T-Lock)
$D^{\text{repo}}(s, T)$	Discount factor from T back to s , repo curve
$D^{\text{T-Lock}}(s, T)$	Discount factor from T back to s , bond zero-yield curve
D_{tT}	Risk-free zero-coupon bond from t to T (Part II)
$r^{\text{repo}}, r^{\text{fun}}$	Repo rate / unsecured funding rate
RF_t	Risk factor: $-\partial P_t / \partial y$ at $y = \text{IRR}_t$
$\text{PV01}(y_f)$	Centred-difference forward-yield risk factor, $B(y_f + \frac{1}{2}\text{bp}) - B(y_f - \frac{1}{2}\text{bp})$
a	Sign convention: $a = +1$ long, $a = -1$ short

A note on conventions. The anonymous practitioner paper (Part I) uses simple compounding inside the bond forward formula and a centred finite-difference PV01. Pucci (Part II) uses the continuous-compounding bond price

$$P_t(y) = e^{-(t_n-t)y} + c \sum_{t_i > t} \alpha_i e^{-(t_i-t)y} \quad (2)$$

to derive analytical greeks. We keep the two conventions in their respective Parts (this matches the source) and reconcile them in section 14.

Part I — Practitioner Pricing Model

3 The Bond Forward Pricing Building Block

The pricing of every T-Lock NPV formula in this Part reduces to the *forward clean price* of the underlying bond at delivery. Ignoring bid-offer (the model is mid-market), the bond forward clean price is¹

$$B_f = \frac{P_0 + A_0}{D^{\text{repo}}(T_0, T_{\text{del}})} - \sum_{i=1}^N \frac{C_i}{D^{\text{repo}}(T_i, T_{\text{del}})} - A_{\text{del}}, \quad (3)$$

where

- P_0 = clean price of the underlying bond at spot;
- A_0, A_{del} = accrued interest on a unit of the bond at T_0 and at T_{del} ;
- C_i = the i -th coupon paid at T_i , $T_0 < T_i \leq T_{\text{del}}$;
- $D^{\text{repo}}(s, T)$ = discount factor from T back to s , retrieved from the repo zero curve;
- N = number of coupons paid in $(T_0, T_{\text{del}}]$.

Pricing parameter BOND_FROM_QUOTE. By default this flag is `true`: $P_0 = \text{MarketQuote}$ if available; else P_0 is derived from yield via the price function $P_0(y)$. This affects sensitivity computation (whether P_0 moves with bond shifts).

The forward yield y_f is implicitly defined by

$$P_{T_{\text{del}}}(y_f) = B_f, \quad (4)$$

where $P_{T_{\text{del}}}(y)$ is the bond's price-yield function evaluated *as if it settles on* T_{del} .

4 The Three Lock Conventions

A T-Lock can be struck on any of three quantities. Throughout this section the present value is for the *investor / buyer / long*.

4.1 Lock the clean price ($P^{\text{T-Lock}}$ struck)

$$\text{NPV}^{T-\text{Lock}} = N \cdot (P^{\text{T-Lock}} - B_f) \cdot D^{\text{T-Lock}}(t, T_{\text{del}}) \quad (5)$$

where

- N = quantity of the underlying bond;
- $P^{\text{T-Lock}}$ = locked clean price of the underlying bond (the strike);
- B_f = bond forward clean price, eq. (3);
- $D^{\text{T-Lock}}(t, T_{\text{del}})$ = discount factor from T_{del} back to valuation date t , retrieved from the *bond zero-yield curve*.

The two-curve structure is essential: B_f is built on the repo curve (carry of the security), while the final payoff is discounted on the bond zero-yield curve (the relevant funding for the cash payoff).

¹This is the standard cash-and-carry no-arbitrage identity: buy the bond spot for $P_0 + A_0$, finance at the repo rate, collect intermediate coupons, deliver at T_{del} .

4.2 Lock the yield ($y^{\text{T-Lock}}$ struck)

$$\boxed{\text{NPV}^{T-\text{Lock}} = M \cdot (y_f - y^{\text{T-Lock}}) \cdot \text{PV01}(y_f) \cdot D^{\text{T-Lock}}(t, T_{\text{del}})} \quad (6)$$

where

- M = trade amount for a yield lock;
- y_f = forward bond yield, eq. (4);
- $y^{\text{T-Lock}}$ = locked bond yield (the strike);
- $\text{PV01}(y_f)$ is the *centred-difference forward-yield risk factor*:

$$\text{PV01}(y_f) = B(y_f + 0.5\text{bp}) - B(y_f - 0.5\text{bp}), \quad (7)$$

i.e. the price change of a unit bond per 1bp shift of y_f ;

- $D^{\text{T-Lock}}(t, T_{\text{del}})$ as above.

Remark 1. $\text{PV01}(y_f)$ is *not* the conventional spot PV01. It is the risk-factor of the *forward-settled* bond, evaluated at the forward yield y_f . Confusing the two is a frequent implementation bug.

4.3 Lock the dirty price

The dirty price is the sum of the clean price and accrued interest for a unit bond. The model first computes the implied clean price strike

$$P^{\text{T-Lock}} = \text{LockedDirtyPrice} - A_{\text{del}}$$

and then applies eq. (5). No new formula is required.

5 Risks and Sensitivities (Practitioner View)

The anonymous note explicitly cautions:

There is no specific indicator of risk sensitivity for T-Locks. Both PV01 and NDELTA are underlier-related. We apply them here as approximations of the T-Lock sensitivity. Therefore we consider it a limitation to apply them as a sensitivity indicator for T-Locks.

The natural risk sensitivities the desk reports are therefore:

- PV01 of the *underlying bond*, multiplied by quantity and discount factor;
- NDELTA (n th-order delta) of the underlying bond, similarly scaled;
- Repo-bucket sensitivity, since eq. (3) depends on D^{repo} at multiple coupon nodes.

Part II will sharpen these into formal greeks of the T-Lock itself and explain why the practitioner proxy is an *overhedge*, not a sensitivity match.

Part II — Hedging and Convexity Analysis

6 The Arbitrage-Free Pricing Formula

Recall the spot price-yield identity:

$$P^{\text{mkt}} = P(\text{IRR}), \quad (8)$$

i.e., the IRR is the yield that reprices the bond.

The T-Lock prescribes the compounded-interest formula

$$P_t(y) = \prod_{i=1}^n \frac{1}{1 + \alpha_i y} + c \sum_{i=1}^n \alpha_i \prod_{j=1}^i \frac{1}{1 + \alpha_j y}, \quad (9)$$

with coupon rate c , issue date $t_0 \geq t$, coupon dates $t_1 < \dots < t_n$ and accrual factors $\alpha_1, \dots, \alpha_n$. When t lies in a coupon period $(t_0, t_1]$ the formula generalises by multiplying with the stub discount $\left(\frac{1}{1 + \alpha_1 y}\right)^{(t_1 - t)/(t_1 - t_0)}$.

With $\text{RF}_t = -\partial P_t / \partial y|_{y=\text{IRR}_t}$, the T-Lock payoff at t_e is g in eq. (1).

First-order interpretation. At first order in $(\text{IRR}_{t_e} - L)$:

$$\begin{aligned} g &= -a \cdot \frac{\partial P_{t_e}(y)}{\partial y} \Big|_{y=\text{IRR}_{t_e}} \cdot (\text{IRR}_{t_e} - L) \\ &\sim a \cdot (P_{t_e}(L) - P_{t_e}(\text{IRR}_{t_e})) = a \cdot (P_{t_e}(L) - P_{t_e}^{\text{mkt}}). \end{aligned} \quad (10)$$

The T-Lock thus measures the gap between the market price of the bond at expiry and its price with cash-flows discounted at the locked rate L . We will show this identifies an *overhedge* for $a = +1$ and an *underhedge* for $a = -1$.

General formula. In an arbitrage-free framework

$$v_t = a \cdot D_{tt_e} \cdot \mathbb{E}_t^{t_e} [\text{RF}_{t_e} \cdot (\text{IRR}_{t_e} - L)], \quad (11)$$

where the expectation is under the t_e -forward measure and D_{tt_e} is the risk-free t_e -zero-coupon bond seen at t . Default risk is ignored: short tenor + US-government issuer.

Why this is not just a CMT swaplet. If RF_{t_e} is *frozen*, the payoff superficially resembles a CMS-let on the bond curve (a Constant Maturity Treasury swaplet, cf. Pucci 2014, SSRN 3387961). This interpretation is *misleading* because freezing RF overshadows — and actually reverses — a key feature: while a CMS-let has positive convexity, the T-Lock has *negative* convexity locally near L (see section 8.4).

7 The Forward-Contract Proxy and Overhedge

7.1 The proxy strike

For ease of analytics, from now on we use the continuous-compounding bond price eq. (2):

$$P_t(y) = e^{-(t_n - t)y} + c \sum_{t_i > t} \alpha_i e^{-(t_i - t)y}.$$

From eq. (10), the payoff of a long Standard T-Lock struck at L is approximated, at first order, by the t -value of a *short* forward contract struck at $P_{t_e}(L)$. The proxy strike is

$$K = P_{t_e}(L) \quad (12)$$

$$\sim \text{ForwardPrice}_t(t_e) + \text{RF}_{t_e} \cdot (\text{IRR}_t - L), \quad (13)$$

where IRR_t is the t_e -forward IRR defined implicitly by $P_{t_e}(\text{IRR}_t) = \text{ForwardPrice}_t(t_e)$.

The forward proxy is a convenient booking representation when the front office system does not feature a T-Lock typology.

7.2 The overhedge error is the Taylor remainder

Substituting P by its first-order Taylor polynomial about L , eq. (10) approximation becomes an exact overhedge once we identify it with the Taylor remainder.

Proposition 1 (Overhedge bound). *Let*

$$R_1(y) = P(y) - P(L) - D_y[P](L) \cdot (y - L).$$

Then R_1 has a unique stationary point at $y = L$ which is also a maximum (since the second derivative of R_1 is negative everywhere). Moreover, by the Remainder Estimation Theorem,

$$|R_1(y)| \leq \frac{M |y - L|^2}{2}, \quad (14)$$

where $M = \max_{y \in I} |D_y^{(2)}[P](y)| = D_y^{(2)}[P](\min I)$ (the maximum is attained at the left-most point of the interval I because $D_y^{(3)}[P] < 0$).

Proof. $D_y^{(2)}[P] > 0$ (positive convexity, eq. (16) below), so R_1 is concave on I . Its first derivative $D_y[R_1](y) = D_y[P](y) - D_y[P](L)$ vanishes only at $y = L$, giving the unique critical point. Concavity makes it a maximum. The remainder bound is the standard Taylor result with $|D_y^{(2)}[P]|$ maximised on I , attained at the left endpoint because $D_y^{(3)}[P] < 0$. $\square$

Numerical magnitude. For the 10Y benchmark

$$M \sim (t_n - t)^2 + c \sum_{t_i > t} (t_i - t)^2 \alpha_i \sim 150$$

(taking the extreme case $y = 0$; minimum 1Y CMT yield was 0.08% in September 2011, so this is a hard upper bound). A 30% yield move — an upper bound for a 6M tenor — yields a dollar error of $\approx 0.005\%$ of notional. The overhedge is therefore tight.

8 Greeks of the T-Lock

8.1 Derivatives of the bond price

From the continuous-compounding form eq. (2):

$$D_y[P_t] = -(t_n - t) e^{-(t_n - t)y} - c \sum_{t_i > t} \alpha_i (t_i - t) e^{-(t_i - t)y} < 0, \quad (15)$$

$$D_y^{(2)}[P_t] = (t_n - t)^2 e^{-(t_n - t)y} + c \sum_{t_i > t} \alpha_i (t_i - t)^2 e^{-(t_i - t)y} > 0, \quad (16)$$

$$D_y^{(3)}[P_t] = -(t_n - t)^3 e^{-(t_n - t)y} - c \sum_{t_i > t} \alpha_i (t_i - t)^3 e^{-(t_i - t)y} < 0. \quad (17)$$

The convexity of the bond is $D_y^{(2)}[P]/P > 0$.

8.2 Payoff derivatives

Writing the payoff as a function of yield, $g(y) = -a D_y[P](y - L)$,

$$D_y[g] = -a [D_y^{(2)}[P] \cdot (y - L) + D_y[P]], \quad (18)$$

$$D_y^{(2)}[g] = -a [D_y^{(3)}[P] \cdot (y - L) + 2D_y^{(2)}[P]]. \quad (19)$$

8.3 Delta of the T-Lock

For $a = +1$ and $|y - L|$ small enough (which is the typical business case: ATM strike, short tenor),

$$D_y[g] > 0. \quad (20)$$

The delta with respect to the security price is then

$$D_P[g] = D_y[g] \cdot D_P[y] < 0, \quad (21)$$

consistent with a long T-Lock being equivalent to shorting the security.

A sharp characterisation: solving eq. (18) for L ,

$$D_y[g] > 0 \text{ and } \Delta < 0 \iff L > y + \frac{D_y[P]}{D_y^{(2)}[P]}. \quad (22)$$

Equivalently $y - L < -D_y[P]/D_y^{(2)}[P] > 0$: as long as y does not overshoot L , Δ stays negative.

Hedging. Accounting for discounting, a trader who books the T-Lock short via the proxy forward of section 7 can delta-hedge by *shorting* an amount of bond equal to

$$\Delta_{\text{hedge}} \sim D_{tt_e} \cdot D_P[g]. \quad (23)$$

The hedge is continuously rebalanced.

8.4 Gamma of the T-Lock

From the second-derivative chain rule (see appendix A, eq. (44)):

$$\begin{aligned} D_P^{(2)}[g] &= D_y^{(2)}[g] \cdot (D_P[y])^2 + D_y[g] \cdot D_P^{(2)}[y] \\ &= D_y^{(2)}[g] \cdot (1/D_y[P])^2 - D_y[g] \cdot [D_y^{(2)}[P]/(D_y[P])^3]. \end{aligned} \quad (24)$$

Substituting eqs. (18) and (19) into eq. (24), for $|y - L|$ small:

$$\begin{aligned} D_P^{(2)}[g] &\sim -a[2 D_y^{(2)}[P]](1/D_y[P])^2 + a[D_y[P]] [D_y^{(2)}[P]/(D_y[P])^3] \\ &= -a[2 D_y^{(2)}[P]](1/D_y[P])^2 + a[D_y^{(2)}[P]/(D_y[P])^2] \\ &= -a \cdot D_y^{(2)}[P] \cdot (1/D_y[P])^2. \end{aligned} \quad (25)$$

For $a = +1$ and $|y - L|$ small, $D_y^{(2)}[P] > 0$ and $(D_y[P])^{-2} > 0$, so

$$\boxed{D_P^{(2)}[g] < 0 \quad (\text{negative gamma / negative convexity}).}$$

Practical consequence. Dynamically replicating a long T-Lock under no-volatility assumption is an *overhedge* generating non-negative P&L for the trader (cf. Carr-Madan 2002 [14, Eq. 9]). Equivalently: delta-hedging a *short* T-Lock position generates *gamma profits*.

8.5 Sharp negativity bounds

A non-local characterisation: from eq. (24),

$$\Gamma < 0 \iff (y - L) \cdot \left[(D_y^{(2)}[P])^2 - D_y[P] \cdot D_y^{(3)}[P] \right] > D_y[P] \cdot D_y^{(2)}[P]. \quad (26)$$

The bracket on the LHS is strictly negative:

Lemma 1. $(D_y^{(2)}[P])^2 - D_y[P] \cdot D_y^{(3)}[P] < 0$.

The proof is deferred to appendix C (eq. (51)). Given the lemma, dividing eq. (26) by the (negative) bracket flips the inequality:

$$\Gamma < 0 \iff y - L < \frac{D_y[P] \cdot D_y^{(2)}[P]}{(D_y^{(2)}[P])^2 - D_y[P] \cdot D_y^{(3)}[P]} > 0. \quad (27)$$

Again: as long as y does not overshoot L , Γ stays negative. Equation (27) gives an explicit **negativity-of-gamma upper bound** we will estimate empirically in section 11.

9 Replication via Repo

The bond underlying the proxy forward is most often shorted via a *reverse repo* trade. Equation (21) gives the delta with respect to the forward price, which is interpreted as the size of the repo to trade.

The forward price relation (zero-haircut) is

$$\boxed{\text{ForwardPrice}_t(t_e) = P_t^{\text{mkt}}(1 + r_{tt_e}^{\text{repo}} \cdot (t_e - t)) - c \sum_{t < t_i \leq t_e} \alpha_i (1 + r_{t_i t_e}^{\text{repo}} \cdot (t_e - t_i)).} \quad (28)$$

This is the simple-compounding equivalent of eq. (3). We assume zero haircut throughout: a good approximation for Treasuries.

Repo-rate sensitivity. By the chain rule on eq. (28) and eq. (13),

$$\frac{\partial g(r^{\text{repo}})}{\partial r^{\text{repo}}} = \frac{\partial g(y)}{\partial y} \Big|_{y=\text{IRR}_t} \cdot \frac{\partial y(P)}{\partial P} \Big|_{P=\text{ForwardPrice}_t} \cdot \frac{\partial \text{ForwardPrice}(r^{\text{repo}})}{\partial r^{\text{repo}}}. \quad (29)$$

A higher repo rate $\Rightarrow$ higher forward price $\Rightarrow$ lower forward yield $\Rightarrow$ *cheaper* long T-Lock position.

Specialness. For benchmark Treasuries trading "on special", the actual repo rate may be significantly below GC. Historical extreme: during a liquidity squeeze the 10Y repo dropped to -300bp (Leong 2014). The fair value of the long T-Lock rises accordingly.

EUR aside. Negative EUR repo rates make a hypothetical T-Lock on a European benchmark more expensive than a USD one.

Maturity	Bid	Ask
1M	2.57%	2.52%
3M	2.60%	2.55%
6M	2.63%	2.58%
12M	2.70%	2.65%

Table: GC repo rates for the 10Y as of 24 Jan 2019.

10 The Then-Current T-Lock and Roll P&L

In a Then-Current T-Lock the underlying is determined at t_e as the on-the-run benchmark prevailing at t_e . At inception this security is unknown / unlisted in the position-keeping system.

Booking convention. The trader books the trade as a *Standard T-Lock with the currently on-the-run security*. At each benchmark roll date t_{roll} , the booking is revised: the underlying is switched to the new on-the-run, generating a *roll P&L*.

10.1 The roll P&L formula

To keep notation light, assume the security is switched from a bond with coupon c to a bond with coupon $\hat{c}$, *all other details equal*. The NPV jumps from

$$\pi_t = D_{tt_e} [K - F_t(t_e)], \quad (30)$$

$$\hat{\pi}_t = D_{tt_e} [\hat{K} - \hat{F}_t(t_e)], \quad (31)$$

with $K = P_{t_e}(L)$, $\hat{K} = \hat{P}_{t_e}(L)$, and the new price-yield curve

$$\hat{P}_t(y) = e^{-(t_n-t)y} + \hat{c} \sum_{t_i > t} \alpha_i e^{-(t_i-t)y}. \quad (32)$$

The P&L due to the switch is

$$\hat{\pi}_t - \pi_t = D_{tt_e} [(\hat{K} - K) - (\hat{F}_t(t_e) - F_t(t_e))]. \quad (33)$$

The strike gap is purely coupon-driven:

$$\hat{K} - K = (\hat{c} - c) \sum_{t_i > t_e} \alpha_i e^{-(t_i - t_e)L}. \quad (34)$$

By definition $F_t(t_e) = P_{t_e}(R_t)$ and $\hat{F}_t(t_e) = \hat{P}_{t_e}(\hat{R}_t)$, so

$$\begin{aligned} \hat{F}_t - F_t &= \hat{P}_{t_e}(\hat{R}_t) - P_{t_e}(R_t) \\ &= [\hat{P}_{t_e}(\hat{R}_t) - P_{t_e}(\hat{R}_t)] + [P_{t_e}(\hat{R}_t) - P_{t_e}(R_t)] \\ &= (\hat{c} - c) \sum_{t_i > t_e} \alpha_i e^{-(t_i - t_e)\hat{R}_t} + [P_{t_e}(\hat{R}_t) - P_{t_e}(R_t)]. \end{aligned} \quad (35)$$

Short-tenor approximation. When $t_n - t_e$ is small (the bond at t_e is close to maturity), the last bracket is linearisable:

$$P_{t_e}(\hat{R}_t) - P_{t_e}(R_t) \sim \left[(t_n - t_e) + c \sum_{t_i > t_e} \alpha_i (t_i - t_e) \right] (R_t - \hat{R}_t). \quad (36)$$

Combining eqs. (34) to (36) into eq. (33), the (forward) P&L satisfies

$$\boxed{\begin{aligned} \frac{\hat{\pi}_t - \pi_t}{D_{tt_e}} &\sim (\hat{c} - c)(\hat{R}_t - L) \sum_{t_i > t_e} \alpha_i (t_i - t_e) \\ &\quad + (\hat{R}_t - R_t) \left[(t_n - t_e) + c \sum_{t_i > t_e} \alpha_i (t_i - t_e) \right]. \end{aligned}} \quad (37)$$

A symmetric expansion gives

$$\frac{\hat{\pi}_t - \pi_t}{D_{tt_e}} \sim (c - \hat{c})(L - R_t) \sum_{t_i > t_e} \alpha_i (t_i - t_e) + (\hat{R}_t - R_t) \left[(t_n - t_e) + \hat{c} \sum_{t_i > t_e} \alpha_i (t_i - t_e) \right]. \quad (38)$$

Hence the security change generates little forward-contract P&L if

1. $t_n - t_e$ is small (close to bond maturity),
2. L is close to R_t ,
3. there is no significant IRR gap between the two securities.

General (longer-tenor) version. Setting

$$A_t(y) \equiv \sum_{t_i > t} \alpha_i e^{-(t_i - t)y},$$

removing the "close-to-maturity" restriction yields

$$\boxed{\frac{\hat{\pi}_t - \pi_t}{D_{tt_e}} \sim (\hat{c} - c) \cdot D_y[A_{t_e}]|_{y=\hat{R}_t} \cdot (L - \hat{R}_t) + D_y[P_{t_e}]|_{y=\hat{R}_t} \cdot (R_t - \hat{R}_t),} \quad (39)$$

with

$$D_y[A_t] = - \sum_{t_i > t} \alpha_i (t_i - t) e^{-(t_i - t)y}. \quad (40)$$

10.2 Forming the new benchmark

US Treasury benchmarks are reissued periodically by auction. The auction process for a roll proceeds:

1. Evaluate R_t , the IRR of the current benchmark.
2. Locate the prospected coupon $\hat{c}$ so the new issue is at par: $P_t^{(\hat{c})}(R_t) = 1$.
3. The auction filters market appetite, producing a final coupon $\hat{c}'$ (possibly $\neq \hat{c}$).
4. Imply the new IRR from $P_t^{(\hat{c}')}(\hat{R}_t) = 1$.

If the market confirms "at the old level" we land at $\hat{R}_t = R_t$, which by eq. (39) minimises the roll P&L.

10.3 Strike gap for the Then-Current proxy

If at expiry the new benchmark coincides with the issue date, the proxy forward strike at inception is

$$K = 1 + \text{RF}_{t_e} \cdot (R_t - L), \quad (41)$$

where RF_{t_e} is computed on the *proxy* security. At t_{roll} the strike gap becomes

$$\begin{aligned} \text{StrikeGap} &= \text{RF}_{t_e}^{\text{new}} \cdot (\hat{R}_{t_{\text{roll}}} - L) - \text{RF}_{t_e}^{\text{old}} \cdot (R_{t_{\text{roll}}} - L) \\ &= (\text{RF}^{\text{new}} - \text{RF}^{\text{old}}) \cdot (\hat{R}_{t_{\text{roll}}} - L) - \text{RF}^{\text{old}} \cdot (R_{t_{\text{roll}}} - \hat{R}_{t_{\text{roll}}}). \end{aligned} \quad (42)$$

11 Statistical Validation (Pucci 2019)

These tables are reproduced verbatim from Pucci 2019. They define the empirical envelopes against which a pricing library can be benchmarked.

11.1 IRR descriptive statistics (since Jan-2008)

IRR	2Y	5Y	10Y
Average	0.88%	1.61%	2.40%
StDev	0.75%	0.61%	0.52%
5th pct	0.24%	0.70%	1.62%
95th pct	2.59%	2.77%	3.34%

11.2 Overhedge error (6M T-Lock on 10Y)

The locked yield L is set at inception IRR, so $y - L$ is the 6M IRR performance $R_{t+6M} - R_t$.

6M	$y - L$	$(y - L)^2$
Average	-0.090%	0.004%
StDev	0.663%	0.006%
5th pct	-1.44%	3.92×10^{-8}
95th pct	0.93%	0.0206%

With $M \sim 150$ from proposition 1: average overhedge error $\sim 0.32\%$ of notional, 95th-percentile bound $\sim 1.55\%$ (materialised in the 2011 US-downgrade episode).

11.3 Upper bound for negative gamma — eq. (27)

Upper bound on $y - L$	2Y	5Y	10Y
Average	22.47%	21.31%	11.44%
StDev	15.20%	0.12%	0.10%
5th pct	11.15%	21.11%	11.28%
95th pct	25.57%	21.51%	11.61%

A move of $> 11\%$ in the 10Y yield within a typical T-Lock tenor ATM has never been recorded.

11.4 Roll P&L on the 10Y (trader's view, 3% locked yield)

10Y Roll P&L	1M	3M	6M	12M
Average	−0.1085%	−0.1069%	−0.1045%	−0.0996%
StDev	0.5028%	0.4953%	0.4841%	0.4615%
5th pct	0.4749%	0.4677%	0.4568%	0.4352%
95th pct	−1.0395%	−1.0238%	−1.0005%	−0.9530%

11.5 Roll P&L on the 5Y (trader's view, 3%)

5Y Roll P&L	1M	3M	6M	12M
Average	−0.05%	−0.04%	−0.04%	−0.04%
StDev	0.22%	0.22%	0.21%	0.18%
5th pct	0.24%	0.23%	0.22%	0.20%
95th pct	−0.43%	−0.42%	−0.40%	−0.35%

11.6 Roll P&L on the 2Y (trader's view, 3%)

2Y Roll P&L	1M	3M	6M	12M
Average	−0.07%	−0.07%	−0.06%	−0.05%
StDev	0.51%	0.51%	0.50%	0.50%
5th pct	0.32%	0.31%	0.31%	0.30%
95th pct	−0.52%	−0.51%	−0.49%	−0.45%

12 Valuation Adjustments

A T-Lock position carries the standard XVA components:

CVA / DVA. Counterparty and own-default risk. Computation is run-of-the-mill, but a short T-Lock is subject to *systemic wrong-way counterparty risk*: in distressed corporate-market scenarios, a flight-to-quality lowers Treasury yields exactly when the corporate client is most exposed, raising the bank's exposure to that client. Pucci argues this may be (partially) offset by a specular increase in DVA — it is hard to envision a regime where the corporate sector suffers and banks thrive.

FVA. The natural hedge is a repo trade. Funding cost arises either from the cash leg the bank posts to borrow the security, or from the collateralisation mechanism if the repo is CSA-assisted or cleared.

KVA. Capital costs are minimal — one of the reasons T-Locks are profitable for the dealer bank, though by the same token mark-ups must remain low.

Proxy underestimation. Because the forward proxy is an overhedge, it *underestimates* the exposure of the short T-Lock, hence underestimates the "contra" contributions (CVA, FVA,

KVA). The short tenor mitigates the consequences, since XVAs for ATM short trades are normally a fraction of the carried exposure.

13 Conclusions (Pucci 2019)

1. A long Standard T-Lock has negative convexity locally near L .
2. A trader can safely book a short T-Lock as a long proxy forward contract on the same security with the same expiry. The forward is a natural overhedge, generating a *conservative* P&L representation.
3. Delta hedging the position generates non-negative *gamma profits*.
4. A Then-Current T-Lock can be booked as a proxy forward on the current on-the-run, with revision at each roll date generating statistically negligible P&L.

14 Validation and Numerical Examples

This section is the bridge to a pricing-library implementation.

14.1 Pucci’s worked example — 24 January 2019

Use this verbatim as a regression test.

Parameter	Value
Trade date t	24 January 2019
Underlying	US Treasury, ISIN US9128285M81
Coupon rate c	3.125%
Maturity	25 November 2028
Expiry t_e	24 April 2019 (3-month tenor)
Locked yield L	2.717% (ATM)
Spot IRR $_t$	2.717%
Spot dirty price P_t^{mkt}	104.1055%
Spot clean price	103.4922%
3M repo rate r^{repo}	2.46%
Forward IRR (IRR $_t$ at t_e)	2.53%
Forward dirty price	104.74%
Forward clean price (B_f)	103.36%

Validation checks. A pricing library implementing the formulas of this document must reproduce, on this dataset:

1. The bond *forward clean price* 103.36% via eq. (3) (Part I) and eq. (28) (Part II). **The two must agree** up to the simple-vs-continuous compounding convention (Part I uses simple, Part II continuous); switch via $D^{\text{repo}} = e^{-r^{\text{repo}}(T_{\text{del}} - T_0)}$ vs $D^{\text{repo}} = 1/(1 + r^{\text{repo}}(T_{\text{del}} - T_0))$.
2. The forward yield $y_f = 2.53\%$ as the inverse of eq. (4): $P_{T_{\text{del}}}(y_f) = B_f$.
3. The T-Lock NPV via all three Part I conventions (eq. (5), eq. (6), dirty-price): at $L = \text{IRR}_t = 2.717\%$, the *spot* NPV is exactly zero (ATM); the forward NPV is non-zero because of carry.
4. The overhedge plot of Figure 1 in Pucci’s paper: payoff $g(y)$ via eq. (1), forward approx via eq. (13), overhedge error via $R_1(y)$ of proposition 1. The error curve is convex with minimum (zero) at $y = L$ and small positive values on either side.

14.2 Library-ready parameter block (pseudocode)

```
# --- Trade & market state ---
trade_date      = 2019-01-24
expiry          = 2019-04-24
isin            = "US9128285M81"
coupon          = 0.03125          # annual
maturity        = 2028-11-25
notional        = 1.0              # 100% of face
locked_yield    = 0.02717

# --- Calibration targets ---
spot_dirty      = 1.041055
spot_clean      = 1.034922
repo_3M         = 0.0246
fwd_irr         = 0.02530
fwd_dirty       = 1.0474
fwd_clean       = 1.0336

# --- Coupon dates in (trade_date, expiry] ---
# (zero or one for a 3M T-Lock on a semi-annual UST; check)
# For 11/25 + 5/25 schedule, no coupon falls in (1/24, 4/24]
```

14.3 Cross-formula consistency

Bond forward, simple vs continuous. Part I writes eq. (3) with simple-compounded discount factors $D^{\text{repo}}(s, T) = 1/(1 + r^{\text{repo}}(T - s))$. Part II writes eq. (28) with continuous-compounded discount. For the 3M example:

- simple: $(1 + 0.0246 \cdot 0.25) = 1.00615$
- continuous: $e^{0.0246 \cdot 0.25} = 1.006159$

identical to 5 decimal places.

PV01(y_f) vs spot PV01. The forward-yield PV01 in eq. (7) uses the bond's price-yield function at *expiry* (settlement = T_{del}). The spot PV01 uses settlement T_0 . The two differ by the carry between T_0 and T_{del} . Library implementation MUST distinguish them (remark 1).

Two-curve discounting in Part I. B_f uses D^{repo} ; the final cash payoff uses $D^{\text{T-Lock}}$ (bond zero-yield curve). These are different curves. In Pucci (Part II), the formal arbitrage-free formula uses a single risk-free $D_{t t_e}$, justified by the short tenor + US-government issuer. The practitioner two-curve approach is the more general (and more careful) representation.

14.4 Risk: what to compute and how to interpret

Delta vs forward price. eq. (21); negative for a long T-Lock at ATM; hedge by shorting the bond via reverse repo.

Gamma vs forward price. eq. (25); negative for $|y - L|$ small, bounded explicitly by eq. (27).

Repo bucket vega. eq. (28) depends on r^{repo} at t_e and at each coupon date $t_i \in (t, t_e]$. Bump each tenor independently for a repo-curve risk decomposition.

PV01 / NDELTA of the underlier. Compute and report as approximations (the anonymous paper's caveat: section 5). They are *not* the sensitivity of the T-Lock itself — those are eqs. (21) and (25).

14.5 Smoke tests

1. **ATM zero NPV.** At $y^{\text{T-Lock}} = \text{IRR}_t$ on the trade date the spot T-Lock NPV is zero (after netting carry). Test all three conventions.
2. **Parallel shift symmetry.** Shift the bond yield by $\pm \Delta y$; payoffs are linear at first order (sign by sign), overhedge error symmetric and quadratic.
3. **Sign convention.** $a = +1$ long, $a = -1$ short; long pays when $\text{IRR} > L$. A long T-Lock + a long bond forward should be approximately delta-neutral.
4. **No-arbitrage with the forward proxy.** $\text{NPV}_{\text{long}}^{\text{T-Lock}} + \text{NPV}^{\text{fwd, short, K=Pte(L)}} = R_1(y)$ remainder, bounded by eq. (14).

A Simple Math: Chain Rule and Inverse Function

For differentiable f, g :

$$(f \circ g)' = f' \cdot g', \quad (43)$$

$$(f \circ g)'' = (f'' \circ g) \cdot (g')^2 + (f' \circ g) \cdot g''. \quad (44)$$

For an invertible differentiable f :

$$(f^{-1})' = \frac{1}{f'}, \quad (45)$$

$$(f^{-1})'' = -\frac{f''}{(f')^3}. \quad (46)$$

Applied to $y = P^{-1}(P)$ (treating P as the independent variable when computing greeks w.r.t. the bond price):

$$D_P[y] = 1/D_y[P], \quad D_P^{(2)}[y] = -D_y^{(2)}[P]/(D_y[P])^3.$$

These are the substitutions used in section 8.4.

B Key Derivatives of the Bond Price

Decomposing $P_t^{(c)}(y) = P_t^{(0)}(y) + c A_t(y)$ with

$$P_t^{(0)}(y) = e^{-(t_n-t)y}, \quad A_t(y) = \sum_{t_i > t} \alpha_i e^{-(t_i-t)y},$$

we have

$$D_y[P_t^{(c)}] = -(t_n - t)e^{-(t_n-t)y} + c D_y[A_t] = D_y[P_t^{(0)}] + c D_y[A_t], \quad (47)$$

$$D_y^{(k)}[P^{(c)}] = (t - t_n)^k e^{-(t_n-t)y} + c \sum_{t_i > t} \alpha_i (t - t_i)^k e^{-(t_i-t)y}, \quad (48)$$

$$D_y^{(k)}[A_t] = \sum_{t_i > t} \alpha_i (t - t_i)^k e^{-(t_i-t)y}, \quad (49)$$

$$D_y^{(k)}[P_t^{(c)}] = D_y^{(k)}[P_t^{(0)}] + c D_y^{(k)}[A_t]. \quad (50)$$

C More Math: The Convexity-Inequality Proof

Throughout, set

$$q_i \equiv (t - t_i), \quad Q_i \equiv \alpha_i e^{-(t_i-t)y}.$$

Lemma 2. $D_y^{(2)}[P_t^{(0)}] D_y^{(2)}[A_t] + D_y^{(2)}[P_t^{(0)}] D_y^{(2)}[A_t] - 2 D_y^{(2)}[P_t^{(0)}] D_y^{(2)}[A_t] \geq 0$.

Proof. After substitution this becomes

$$q_n Q_n \sum_{t_i > t} q_i^3 Q_i + q_n^3 Q_n \sum_{t_i > t} q_i Q_i - 2 q_n^2 Q_n \sum_{t_i > t} q_i Q_i = \sum_{t_i > t} Q_i Q_n q_i q_n [q_i - q_n]^2 \geq 0,$$

clearly non-negative. $\square$

Lemma 3. $(D_y^{(2)}[A_t])^2 - D_y^{(2)}[A_t] \cdot D_y^{(2)}[A_t] < 0$.

Proof. We must show

$$\left[\sum_{t_i > t} q_i Q_i \right] \left[\sum_{t_i > t} q_i^3 Q_i \right] - \left[\sum_{t_i > t} q_i^2 Q_i \right]^2 > 0,$$

since the LHS equals $\sum_{t_i, t_j > t} Q_i Q_j [q_i - q_j]^2 > 0$ by Cauchy-Schwarz. Concretely:

$$\begin{aligned} \sum_i q_i Q_i \sum_j q_j^3 Q_j - \left[\sum_i q_i^2 Q_i \right]^2 &= \sum_{i \neq j} q_i q_j^3 Q_i Q_j - \sum_{i \neq j} q_i^2 q_j^2 Q_i Q_j \\ &= \sum_{i < j} (q_i q_j^3 + q_i^3 q_j - 2 q_i^2 q_j^2) Q_i Q_j \\ &= \sum_{i < j} q_i q_j (q_i - q_j)^2 Q_i Q_j > 0. \end{aligned} \quad \square$$

Proof of lemma 1. We must show $(D_y^{(2)}[P])^2 - D_y[P] \cdot D_y^{(3)}[P] < 0$. Using the decomposition eq. (50), and expanding:

$$\begin{aligned} (D_y^{(2)}[P])^2 &= (D_y^{(2)}[P^{(0)}])^2 + 2c D_y^{(2)}[P^{(0)}] D_y^{(2)}[A] + c^2 (D_y^{(2)}[A])^2, \\ D_y[P] D_y^{(3)}[P] &= D_y[P^{(0)}] D_y^{(3)}[P^{(0)}] + c [D_y[P^{(0)}] D_y^{(3)}[A] + D_y[A] D_y^{(3)}[P^{(0)}]] + c^2 D_y[A] D_y^{(3)}[A]. \end{aligned}$$

The leading $(D_y^{(2)}[P^{(0)}])^2$ and $-D_y[P^{(0)}] D_y^{(3)}[P^{(0)}]$ cancel (both are computed directly from the zero-coupon factor and one verifies that $(D_y^{(2)}[P^{(0)}])^2 = D_y[P^{(0)}] D_y^{(3)}[P^{(0)}]$, both equal to $(t_n - t)^4 e^{-2(t_n - t)y}$). Subtracting and grouping:

$$(D_y^{(2)}[P])^2 - D_y[P] D_y^{(3)}[P] = c \mathcal{L} + c^2 \mathcal{Q}, \quad (51)$$

with

$$\begin{aligned} \mathcal{L} &= 2 D_y^{(2)}[P^{(0)}] D_y^{(2)}[A] - D_y[P^{(0)}] D_y^{(3)}[A] - D_y[A] D_y^{(3)}[P^{(0)}], \\ \mathcal{Q} &= (D_y^{(2)}[A])^2 - D_y[A] D_y^{(3)}[A]. \end{aligned}$$

By lemma 2 $\mathcal{L} \leq 0$ and by lemma 3 $\mathcal{Q} < 0$. With $c > 0$, the whole expression is negative. $\square$

D The Forward Price and the Repo Strategy

A *coupon security* pays a discrete stream of cash flows c_i at times T_i . For a generic haircut $h^{\text{cut}} \in [0, 1]$, the T -forward price under continuous compounding is

$$\boxed{\text{ForwardPrice}_t(T) = X_t \left[(1 - h^{\text{cut}}) e^{r^{\text{repo}}(T-t)} + h^{\text{cut}} e^{r^{\text{fun}}(T-t)} \right] - \sum_{t < T_i \leq T} c_i e^{(h^{\text{cut}} r^{\text{fun}} + (1-h^{\text{cut}}) r^{\text{repo}})(T-T_i)}. \quad (52)$$

Rates are continuously compounded and flat for ease of illustration.

D.1 The repo-cash strategy (zero haircut)

In a repo trade with intermediate coupons, the trader:

1. At t , borrows X_t from the repo counterparty at rate r^{repo} until T ; buys the bond and transfers it to the counterparty as collateral.
2. On each coupon date T_i , receives coupon c_i and decreases the outstanding by c_i .
3. At T , receives the bond back, sells it in the market, and returns all outstanding cash.

Let A_u = repo cash account at time u (negative = owed). At start: $A_t = -X_t$. At T_1 :

$$A_{T_1} = c_1 + A_t e^{r^{\text{repo}}(T_1-t)}.$$

Recursively for $i = 2, \dots, q$:

$$A_{T_i} = c_i + A_{T_{i-1}} e^{r^{\text{repo}}(T_i-T_{i-1})}.$$

Between T_q and T it accrues to

$$A_T = A_{T_q} e^{r^{\text{repo}}(T-T_q)}.$$

Lemma 4. For all $u \leq T$,

$$A_u = \sum_{t < T_i \leq u} c_i e^{r^{\text{repo}}(u-T_i)} - X_t e^{r^{\text{repo}}(u-t)}. \quad (53)$$

Proof. Induction. Base: $A_t = -X_t$. Inductive step from T_{i-1} to T_i :

$$\begin{aligned} A_{T_i} &= c_i + A_{T_{i-1}} e^{r^{\text{repo}}(T_i-T_{i-1})} \\ &= c_i + \left[\sum_{t < T_j \leq T_{i-1}} c_j e^{r^{\text{repo}}(T_{i-1}-T_j)} - X_t e^{r^{\text{repo}}(T_{i-1}-t)} \right] e^{r^{\text{repo}}(T_i-T_{i-1})} \\ &= c_i + \sum_{t < T_j \leq T_{i-1}} c_j e^{r^{\text{repo}}(T_i-T_j)} - X_t e^{r^{\text{repo}}(T_i-t)} \\ &= \sum_{t < T_j \leq T_i} c_j e^{r^{\text{repo}}(T_i-T_j)} - X_t e^{r^{\text{repo}}(T_i-t)}. \end{aligned}$$

Free accrual between T_q and $u \leq T$ preserves the form. □

Forward-price identity. The strategy is zero-cost at t and yields at T :

$$v_T = X_T + A_T.$$

By definition A_T is the T -forward price:

$$\text{ForwardPrice}_t(T) = X_t e^{r^{\text{repo}}(T-t)} - \sum_{t < T_i \leq T} c_i e^{r^{\text{repo}}(T-T_i)}.$$

(54)

This is eq. (52) at $h^{\text{cut}} = 0$.

D.2 Unsecured-funding limit and the blend

If the repo trade is unavailable, all funding is unsecured at rate r^{fun} :

$$\text{ForwardPrice}_t(T) = X_t e^{r^{\text{fun}}(T-t)} - \sum_{t < T_i \leq T} c_i e^{r^{\text{fun}}(T-T_i)}. \quad (55)$$

This is eq. (52) at $h^{\text{cut}} = 1$.

A generic haircut $0 \leq h^{\text{cut}} \leq 1$ produces the blend eq. (52), derivable from a strategy that distributes coupon payments between repo redemption and unsecured-funding redemption according to their relative weights.

For a Treasury we assume $h^{\text{cut}} = 0$ throughout the body of this study.

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